Inertial DK

John Bell[1]

[1]Lawrence Berkeley National Laboratory, Berkeley, California, 94720, USA

Abstract

1 Setting

Simple model for flocking

$$\begin{aligned} dX_i &= v_i dt \\ dv_i &= -\gamma v_i dt - \nabla U(x) dt + \sqrt{2D} dW_t^i \end{aligned}$$

Resulting SPDE from MullerRenesseZimmerman2025 is

$$d\rho = D\Delta_{\theta\theta}\,\rho\,dt - \nabla_x \cdot (v\rho) + \nabla_v \cdot [(\gamma v + v\nabla U)\rho]\; dt + \frac{1}{\sqrt{N}} \nabla_\theta \cdot (\sqrt{2\rho D}\, dB) \tag{1}$$

where N is the number of particles and dB denotes space-angle-time vector-valued white noise.

Treat $\nabla_x \cdot (v\rho)$ term using methodology used in active matter version.

For particles:

In addition to location X_i each particle should have an assigned velocity. For the moment we can move particles and update velocity with simple explicit Euler-maruyama. Later we will want to put in some version of a verlet integrator but i will specify that later.

restrict time step so that particles move at most 1 cell in x and v in a time step most of the time. if step is too large truncate movement to one cell.

We would like this to work in multilevel hybrid form (see multilevel source)

multilevel considerations.

simple first test is whether particles match finite volume version of SPDE

References